

Comparing Models, Qualifying Conclusions

Comparing models and qualifying what the data supports

Directions:

- Two questions run through the whole sheet: *which model describes this data better?* and *what does that entitle me to say?*
- A residual is observed – predicted. Squaring the residuals and adding them gives one number per model, and the smaller number is the closer fit.
- Every claim you write should name the group it applies to. A number computed from a sample describes that sample first, and anything beyond it needs a reason.

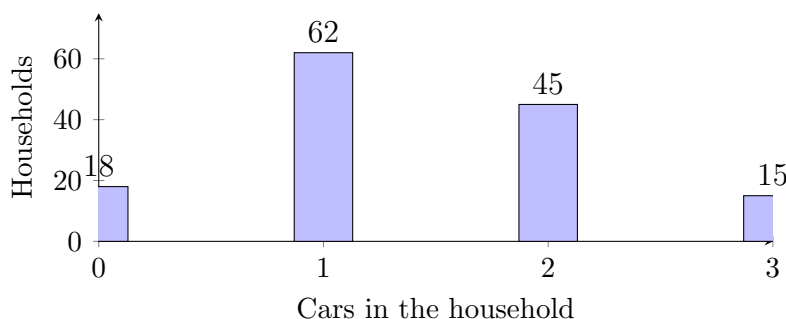
Data set 1 — rainfall. Daily rainfall at one station over nine days, in millimetres: 2, 4, 6, 7, 8, 10, 12, 14, 27. (Problems 1, 2 and 5.)

Data set 2 — plant height. One sunflower measured at the end of each of six weeks, in centimetres, with two candidate straight-line models. (Problems 3, 6, 9, 11 and 12.)

Week x	1	2	3	4	5	6
Height y (cm)	4	7	11	14	18	20

Model A: $\hat{y} = 1.0 + 3.3x$ **Model B:** $\hat{y} = 2.5 + 3.0x$

Chart A — cars per household. A random sample of households in one town. (Problem 8.)



- Data set 1.** A report describes a typical day's rainfall with the *median*. Compute it.

Answer: _____ mm

2. Data set 1. The same report quotes the *interquartile range* as its measure of spread. Compute it.

Answer: _____ mm

3. Data set 2. Find the residual of **Model A** at week 2 — that is, the observed height minus the height Model A predicts.

Answer: _____ cm

4. A survey of 200 students in grades 9 and 10 at one school recorded how each travels to school.

	Bus	Walk	Car	Total
Grade 9	40	30	30	100
Grade 10	25	35	40	100
Total	65	65	70	200

(a) What proportion of the students surveyed walk? Give an exact fraction in lowest terms:

(b) Explain which population this proportion describes, and name one group it should not be extended to.

5. Data set 1. A second report describes the same nine days with the *mean* instead.

(a) Compute the mean.

Answer: _____ mm

(b) The two reports now disagree about a typical day. Explain which summary you would use and why, naming the value in the data that causes the disagreement.

6. Data set 2. The six residuals for **Model A** are

$$-0.3, \quad -0.6, \quad 0.1, \quad -0.2, \quad 0.5, \quad -0.8.$$

Square them and add. This sum of squared residuals is Model A's fit score.

Answer: _____

7. A researcher records the weekly screen time and the reading score of 500 students and finds a strong negative correlation between them. A headline reports that screen time lowers reading scores. Explain why the study cannot support that claim, and name one variable that could produce the correlation without screen time causing anything.

Answer: _____

8. Chart A.

(a) How many households were surveyed altogether? _____

(b) Which number of cars is the modal category? _____

9. Data set 2. The six residuals for **Model B** are

$$-1.5, \quad -1.5, \quad -0.5, \quad -0.5, \quad 0.5, \quad -0.5.$$

(a) Compute Model B's sum of squared residuals: _____

(b) Model A scored 1.39 in problem 6. Explain which model fits the six measured weeks better, and why that comparison does not by itself justify using the winner to predict the height at week 20.

10. Use the travel-survey table from problem 4.

(a) Among grade 9 students, what proportion took the bus? _____

(b) Among grade 10 students, what proportion took the bus? _____

(c) Explain what the comparison of those two proportions does and does not support about grade 9 students in general.

11. Data set 2. Both models are now used to predict the height at week 8, two weeks past the last measurement.

(a) Model A's prediction:

Answer: _____ cm

(b) Model B's prediction:

Answer: _____ cm

12. Synthesis challenge. Data set 2.

(a) By how much do the two week-8 predictions differ?

Answer: _____ cm

(b) Write a two-sentence conclusion about this sunflower's growth that the data genuinely support. The second sentence must be a qualification — a limit on what the models can be trusted to say.